

Forecasting Results

MAE: 1486.77

MAPE: 5.62%

R² Score: 0.925

Inventory Planning

Average Daily Demand: 24,582

Safety Stock: 28,019

Reorder Point: 200,095

Recommendations

- Update forecasts weekly.
- Monitor high demand periods.
- Maintain recommended safety stock levels.
- Reorder inventory before reaching the calculated reorder point.
- Use forecast outputs to optimize purchasing decisions.

Demand Forecasting & Inventory Planning Report

Prepared using Python, Pandas and Prophet

Executive Summary

The business recorded total sales of **47,704,512** units across **10** stores and **50** products. The forecasting model achieved **MAPE of 5.62%** and **R² of 0.925**, indicating strong predictive accuracy. The recommended reorder point is **200,095** units with a safety stock of **28,019** units.

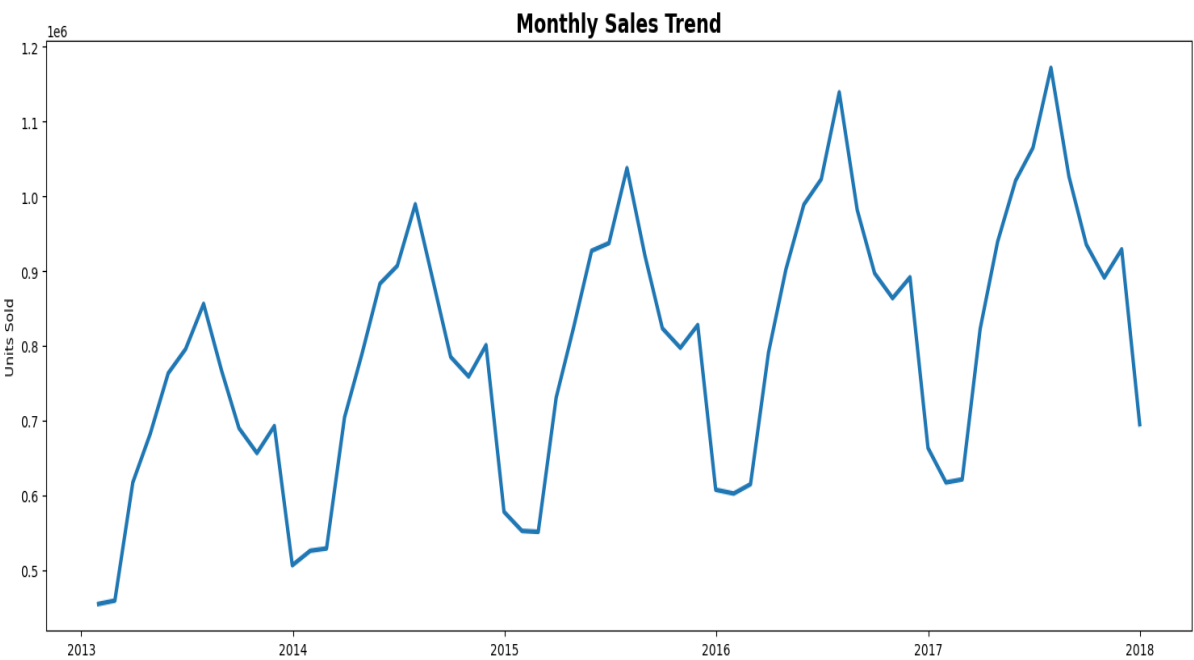
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Business KPIs

Total Units Sold: 47,704,512
Average Daily Sales: 52.25
Stores: 10
Items: 50
Days Covered: 1825

Sales Trend Analysis



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